RESEARCH ARTICLE

Meaning in Life and Depressive Symptoms: A Person-Oriented Approach in Residential and Community-Dwelling Older Adults

Aging and Mental Health

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Objectives: In current society, a significant rise of the population of older adults as well as a high prevalence of depressive symptoms in late life is noticeable. A possible protective resource is 'Meaning in Life'. The objective of this study is to identify from a person-oriented view (a) Meaning in Life-profiles, based on Presence of Meaning and Search for Meaning dimensions, and (b) their associations with depressive symptoms.

Methods: A sample of 205 residential older adults (M = 83.20 years, SD = 7.26) and 280 community-dwelling older adults (M = 75.98 years, SD = 4.76) completed questionnaires of Meaning in Life and depressive symptoms. Firstly, cluster analyses examined potential Meaning in Life-profiles. Secondly, ANOVA tested associations between these distinct profiles and depressive symptoms.

Results: In both samples three distinguishable profiles emerged, a *'Low Presence Low Search'*, a '*High Presence High Search'* and a '*High Presence Low Search'*. Furthermore, older adults with a *High Presence Low Search* profile witnessed less depressive symptoms, compared to those with a *Low Presence Low Search* profile. Residential older adults within the *High Presence High Search* cluster scored in-between the two other clusters for depressive symptoms. However, community-dwelling older adults within this cluster reported similar levels of depressive symptoms as the *High Presence Low Search* group.

Conclusion: Similar Meaning in Life-profiles were detected in residential as well as community-dwelling older adults. In both samples, older adults with a *High Presence Low Search* profile reported less depressive feelings, pointing to the importance of spontaneously experiencing Meaning in Life in this life stage.

Key words: Meaning in Life, depressive symptoms, older adults and cluster analyses

Word count: 4493

Introduction

Current Western society is confronted with a significant rise of the population of older adults. Research estimates that in 2050 10% of the Belgian population will reach the age of 80 or older (Vaes et al., 2010) and the number of older adults aged 65 or more is expected to increase from 18% to 25% between 2013 and 2050 (Federal Plan Office, 2013). A similar, although less pronounced, pattern is expected in the United States with an increase from 12.7% in 2000 to 20.3% in 2050 (Wiener & Tilly, 2002). This extended period of late life may bring several new challenges along, such as chronic disease and bereavement. Research showed a high prevalence of depressive symptoms among older adults, mostly due to age-related increases in risk factors, like isolation or being afraid to fall (van't Veer-Tazelaar et al., 2008). A review by Djernes (2006) pointed to prevalence rates of depressive symptoms fluctuating from 11% to 48% for residential older adults and from 5% to 49% for older adults living in private households. Tsai, Yeh, and Tsai (2005) concluded that community-dwelling older adults, compared to those living in a residential setting, show a lower tendency to develop depressive symptoms. Moreover, the development of depressive symptoms often seems related with negative aspects of functioning such as a decline in daily activities (Metha, Yaffe, and Covinsky, 2002), non-psychiatric hospitalization and even higher mortality rates and suicide risks (Prina, Deeg, Brayne, Beekman, &, Huisman, 2012; Quan, Arboleda-Flórez, Fick, Stuart, & Love, 2002).

Given the prevalence of depressive symptoms in this life stage and their detrimental influence on late life functioning, attention is warranted to identify potential protective factors (Read, Aunola, Feldt, Leinonen, & Ruoppila, 2005). Identification of these strengths might offer tools for policy and health care, and might stimulate the search for effective paths to successful aging (Lewis, 1997).

Meaning in Life

A concept receiving increasing interest as potentially important for optimal functioning is Personal Meaning or Meaning in Life (Dezutter et al., 2013; Dezutter et al., 2014; Steger, Mann, Michels, & Cooper, 2009). Interest in this concept knows a long history (Steger, 2012), however especially the Existential Theory of Victor Frankl (1968) placed the concept of meaning on the map of psychology and psychiatry. Frankl considers meaning essential for healthy coping: Individuals have a primary motivational force to search for meaning and when they are not capable of forming this sense of meaning, psychological distress arises.

Nowadays, several researchers agree that meaning is a multidimensional phenomenon with multiple sources (e.g., Chamberlain & Zika, 1988). Recently, Steger, Frazier, Oishi, and Kaler (2006) further elaborated on the concept and labeled it ‘Meaning in Life’. They define this as ‘the sense made of, and significance felt regarding, the nature of one’s being and existence’ (Steger et al., 2006, p. 81). Steger and colleagues distinguished two dimensions, ‘Presence of Meaning’ and ‘Search for Meaning’ (Steger et al., 2006). Presence of Meaning refers to ‘the subjective sense that one’s life is meaningful’ (Steger et al., 2006, p. 85). It implies some kind of life evaluation and is a highly desired psychological quality. Search of Meaning is ‘the drive and orientation toward finding meaning in one’s life’ (Steger et al., 2006, p. 85) or ‘the strength, intensity, and activity of people’s desire and efforts to establish and/or augment their understanding of the meaning, significance, and purpose of their lives’ (Steger, Kashdan, Sullivan, & Lorentz, 2008, p. 200). It reflects a more dynamic, process-oriented quality. Opinions regarding the desirability of this component differ. Frankl (1968), for example, assumed that searching is a sign of mental health, whereas others perceived it as symptomatic for ill-health (e.g., Klinger, 2012). Reker (2000) assumed there are both healthy and unhealthy forms of Search for Meaning. The nature of this form depends on the motivational origin for the search of meaning (see Steger et al., 2008).

Meaning in Life and Psychological Functioning

Most research showed that Presence of Meaning is associated with benefits, like positive emotions (Steger et al., 2006), life satisfaction and happiness (Steger, Kawabata, Shimai, & Otake, 2008; Steger, Oishi, & Kesebir, 2011), even independent of age (Steger et al., 2009). Other studies revealed negative correlations with indicators of psychological distress, for example negative emotions (e.g., fear, shame and sadness), and on a symptomatic level with depressive symptoms and anxiety symptoms (Debats, 1996; Steger et al., 2006; Steger et al., 2009). A recent study by Scrignaro et al. (2012) showed that Presence of Meaning is related with well-being among cancer patients. Furthermore, it revealed that within this same population, Search for Meaning was negatively correlated with this variable. However, this relation was mediated by the Presence component. The researcher therefore point to the importance of clarifying the relation between Presence of Meaning and Search for Meaning. But, in general, the role of Search for Meaning remains less examined and available findings lack clarity. Some studies indicated that Search for Meaning is related to lower well-being (e.g., Dezutter et al., 2014; Schwartz et al., 2011) and other studies show mixed results. For example, Steger et al. (2008) found a positive correlation between Search for Meaning and rumination and depression, but also between Search for Meaning and open-mindedness and curiosity.

It is remarkable that few studies focused on late life when investigating the role of Presence of Meaning and Search for Meaning (see Steger, 2012). Especially because the few available studies suggested that meaning is an important issue in late life. For example, Reker (1997) showed that a lack of meaning is a predictor for depression in residential older adults. Furthermore, Reker and Woo (2011) found that when community-dwelling older adults find a sense of meaning, they show lower levels of clinical depression. Given the increased prevalence of depressive symptoms, the significant rise of the older population and the possible protective role of meaning in life, more insight is necessary in how Presence of Meaning and Search for Meaning may relate to depressive symptoms in late life.

Person-oriented approach

In variable-oriented research, Presence of Meaning and Search for Meaning are investigated separately, focusing on relationships between and among these constructs (Scholte, Van Lieshout, & de Wit, 2005; Stern, 1911). Recent person-oriented studies, however, pointed to the existence of specific Meaning in Life-profiles, built by combining these components (Dezutter et al., 2013; Dezutter et al., 2014). Thus, instead of classifying variables, this approach classifies individuals and reveals intra-individual profiles (Asendorpf, 2002). Moreover, these Meaning in Life-profiles found in earlier studies showed unique associations with external correlates of individual functioning. In sum, a person-oriented approach can contribute to the understanding of the interplay between the dimensions of Meaning in Life and their relation with psychological functioning. Given the importance of meaning-related topics in late life, investigation of the existence of Meaning in Life-profiles in older adults seems warranted.

Current Study

The first objective of this study was to examine whether naturally occurring Meaning in Life-profiles, found in emerging adults and chronically ill adults (Dezutter et al., 2013; Dezutter et al., 2014), also appear in older adults. Due to different prevalence rates of depressive symptoms between residential and community-dwelling older adults (Tsai et al., 2005), this study focused on both groups separately. In line with earlier studies, we expected four profiles to emerge: (a) a cluster consisting of older adults who experience high levels of Presence of Meaning with low levels of Searching for Meaning (‘*High Presence Low Search’*); (b) a cluster with the opposite profile – consisting of older adults who report low levels of Presence of Meaning and high levels of Search for Meaning (‘*Low Presence High Search’*); (c) a cluster consisting of older adults with high scores both on Presence of Meaning and Search for Meaning (‘*High Presence High Search’*); and (d) a cluster consisting of older adults scoring low on Presence for Meaning as well as on Search of Meaning (‘*Low Presence Low Search’*).

The second objective was to examine the relation between these distinct Meaning in Life-profiles and depressive symptoms (for an overview, see Table 1). In line with earlier research, we anticipated that clusters characterized by high levels of Presence of Meaning will show lower levels of depressive symptoms compared to clusters with a low level of Presence of Meaning. Furthermore, we hypothesized that high levels of Search for Meaning combined with low levels of Presence of Meaning will be associated with high levels of depressive symptoms, whereas high levels of Search for Meaning combined with high levels of Presence of Meaning will be associated with low scores on depressive symptoms. Earlier studies (Dezutter et al., 2013; Dezutter et al., 2014) revealed that Search for Meaning has distinct consequences for individuals with high versus low Presence of Meaning, that is, if individuals experience meaning and search for it, Presence of Meaning seemed to buffer the negative impact of Searching for Meaning.

Method

Participants and Procedure

Nine psychology students, pursuing a master degree, recruited the participants. Exclusion criteria for participation were: <70 years, geriatric cognitive disorders and severe illness. Each questionnaire was accompanied by a letter that introduced the study as an investigation on successful aging. Furthermore, the letter explained that if an individual is interested in participating in the study, a master student will visit to hand over the questionnaire (independent living older adults) or to read out the questionnaire (residential older adults). Informed consent contained the duration of the participation (30 minutes), pointed out that no wrong answers are possible because we are interested in their personal opinion, states that the answers will be treated confidential and that no identification is necessary on the questionnaire (anonymity in data analyses). Finally, the informed consent stated that every individual can withdraw from participation at any moment.

For the residential sample, the boards of the nursing homes were contacted by the second author to introduce the study and ask for involvement. All nursing homes were located in Flanders, Belgium. The boards of the nursing homes contacted their ethical advisory committee and gave their approval for the study. The questionnaires were administered in the nursing homes by five of the master students, more specifically in the private room of the participant or in the leisure room of the nursing home. The questions were read out aloud by the master student and answers were noted on the questionnaire. A total of 205 questionnaires (response rate 82%) were filled out and participants' mean age in the residential older adults was 83.20 years (*SD* = 6.94, range = 70-103). A breakdown by gender yielded 74% women. In this population, 9% was single, 20% was married, 1% cohabited, 67% was widowed, and 5% was divorced. With regard to educational level, 53% completed primary school, 32% both primary and secondary school and 15% higher education.

The group of community-dwelling older adults was recruited by four other master students. They distributed questionnaires in diverse organizations and communities (e.g., social network of grandparents, older adults' social activity clubs) and 280 questionnaires (response rate 80%) were filled out. In this population the mean age was 75.98 years (*SD* = 4.76, range = 70-91) and a breakdown by gender yielded 56% women. Among community-dwelling older adults, 5% was single, 68% was married, 4% cohabited, 21% was widowed, and 2% was divorced. Concerning educational level, 31% completed primary school, 35% both primary and secondary school and 33% higher education.

Instruments

***Meaning in Life*.** Participants rated the 10 items of the Meaning in Life questionnaire (MLQ; Steger et al., 2006). This questionnaire has two subscales, 'Presence of Meaning' and 'Search for Meaning', each comprising five items. Each item was scored on a 4-point Likert scale, with 1 (*absolutely untrue*) till 4 (*absolutely true*). The Presence-subscale (MLQ-P) focusses on cognitive appraisals of experiencing meaning and purpose in life (Cronbach's alpha = .75 for both samples; e.g., 'I have a good sense of what makes my life meaningful.'). The Search-subscale (MLQ-S) assesses the tendency to actively search for purpose and meaning (Cronbach's alpha = .76 for the residential sample and .85 for the community-dwelling sample; e.g., ‘I am seeking a purpose or mission for my life.’). Evidence for convergent and discriminant validity was found. Confirmatory factor analyses in multiple populations revealed a two factor structure, good internal consistency and good one-month test-retest stability (Steger et al., 2006).

***Depressive Symptoms*.** The Center of Epidemiologic Studies Depression scale (CES-D; Radloff, 1977) was used to measure depressive symptoms. This questionnaire recorded in terms of frequency how often people have experienced a certain symptom during the past week (e.g., ‘During the past week, I felt lonely.’). Participants filled out the CES-D short version. This 10-item version is especially developed for use with older adults (Kohout, Berkman, Evans, & Cornoni-Huntley, 1993). Each item was scored on a 4-point scale going from 1 (*rarely or never*) till 4 (*mostly or always*). The CES-D short version consists of the same symptoms dimensions as the larger form and showed good internal consistency in the present study (Cronbach’s alpha = .77 for the residential sample and.79 for the community-dwelling sample).

Data analysis

Cluster analyses were conducted on standardized *z*-scores of both components of Meaning in Life. Scores were standardized within the total sample In the first step, a hierarchical cluster analysis was carried out, using Ward’s method and squared Euclidian distances (Steinley & Brusco, 2007). In the second step, the cluster centers from this hierarchical analysis were used as nonrandom starting points in a non-iterative *k*-means clustering procedure (Breckenridge, 2000). This two-step procedure remedies one of the major shortcomings of the hierarchical method, namely that once a case is clustered, it cannot be reassigned to another cluster at a subsequent stage. *K*-means clustering minimizes within-cluster variability and maximizes between-cluster variability, allowing reassignments to ‘better fitting’ clusters and thus optimizing cluster membership (Gore, 2000). Next, a set of Analyses of Variance (ANOVA) were undertaken to test cluster differences on depressive symptoms.

Results

Preliminary Results

Table 2 represents the basic statistics and correlations among the three variables in both populations. Only the negative correlation between Presence of Meaning and depressive symptoms was significant in both populations.

Cluster-analyses

In the sample of residential older adults, we considered two to six cluster solutions. We firstly compared these various solutions using the Calinski-Harabasz index (CH; Bouxin, 2005; Steinley, 2006), which indicated that the 3-cluster solution provided the best fit (Table 3). Furthermore, we inspected the percentage of variance in the clustering variables that is explained by the cluster solution (Milligan & Cooper, 1985). The explained variance in Presence of Meaning and Search for Meaning increased by 59% when moving from 2 to 3 clusters, by 14% when moving from 3 to 4 clusters, by 11% when moving from 4 to 5 clusters, and by 10% when moving from 5 to 6 clusters, confirming a 3-cluster solution as the best fitting cluster solution. Figure 1 presents the final cluster solution. Because the clusters were defined using *z*-scores for the total sample, the cluster's mean *z*-scores indicate how far that cluster deviates from the total sample mean score and from the means of the other clusters (Scholte et al., 2005). The distances, in standard-deviation (*SD*) units, among the clusters' means (and between each cluster mean and the total sample mean, which is standardized to zero) may be interpreted as an index of effect size. Analogous to Cohen's *d*, 0.2 *SD* represented a small effect, 0.5 *SD* a moderate effect and 0.8 *SD* a large effect. The found clusters were characterized by *z*-scores reflecting moderate to strong deviations from the overall mean, suggesting that the clusters differed considerably on their scores on Presence of Meaning and Search for Meaning. The Low Presence Low Search cluster (n = 70) consisted of older adults low on both Presence of Meaning and Search for Meaning. The High Presence High Search cluster (n = 54) consisted of those high on both Presence of Meaning and Search for Meaning. The High Presence Low Search cluster (n = 74) consisted of older adults high on Presence of Meaning and low on Search for Meaning.

We conducted a chi-square analysis to examine if the cluster distribution differed by socio-demographic variables. However, no significant differences were found for gender, χ^2 (2) = 0.83, $p = ns$, Cramér's V = .07; or educational level χ^2 (4) = 7.52, $p = ns$, Cramér's V = .14. Age differences were found between the clusters, F(2, 195) = 3.56, p < .05), indicating that older adults in the Low Presence Low Search cluster were, on average, significantly older (M = 84.96, SD = .83) than those in the High Presence High Search cluster (M = 81.78, SD = .94).

To test whether the clusters showed distinguished levels of experienced depressive symptoms, an ANOVA was conducted. Results indicated significantly higher levels of depressive symptoms for the Low Presence Low Search profile (M = 1.87, SD = .64) compared with those in the High Presence Low Search profile (M = 1.44, SD = .36) ($F(2, 195) = 11.95$, $p < .001$, $\eta^2 = .11$). Older adults in the High Presence High Search cluster (M = 1.66, SD = .55) reflected levels of depressive symptoms who were in-between the two other clusters.

Also for the community-dwelling older adults, we considered two to six cluster solutions. The CH index (Bouxin, 2005; Steinley, 2006) (see also Table 3) of these solutions also pointed to the 3-cluster solution as the best-fitting solution, next to a 6-cluster solution. The explained variance in Presence of Meaning and Search for Meaning increased by 55% when moving from 2 to 3 clusters, by 16 % when moving from 3 to 4 clusters, by 13% when moving from 4 to 5 clusters, and by 10% when moving from 5 to 6 clusters, confirming a 3-cluster solution as the best fitting cluster solution. Figure 2 shows the final cluster solution for the community-dwelling older adults. In the clusters of the community-dwelling older adults, z-scores reflected small to strong deviations from the overall sample mean. Similar clusters as in the residential older adults population were found namely, a Low Presence Low Search cluster (n = 112), a High Presence High Search cluster (n = 88) and a High Presence Low Search cluster (n = 50).

Again, we conducted a chi-square analysis to examine the extent to which the cluster distribution differed by socio-demographic variables. Similarly to the residential sample, no significant differences were found for gender, $\chi^2(2) = 1.95$, $p = ns$, Cramér's V = .09; or educational level $\chi^2(4) = 6.50$, $p = ns$, Cramér's V = .17. In addition, no age differences were found between the clusters, $F(2, 225) = 1.54$, p = ns.

To test whether the clusters showed distinguished levels of depressive symptoms, an ANOVA was conducted. In line with the residential older adults, higher levels of depressive symptoms were revealed in the Low Presence Low Search cluster (M = 1.82, SD = .53) in comparison with the High Presence Low Search cluster (M = 1.59, SD = .43) and the High Presence High Search cluster (M = 1.56, SD = .40), $F(2, 245) = 8.85$, $p < .001$, $\eta^2 = .07$. In contrast with the residential older adults, High Presence High Search profiles showed similar levels of depressive symptoms as High Presence Low Search profiles and did not function as an in-between cluster.

Discussion

This study identifies distinctive Meaning in Life-profiles in a population of older adults. Firstly, a population of residential older adults was analyzed. Secondly, this was repeated for community-dwelling older adults. Cluster analyses found three distinct Meaning in Life-profiles in both samples: a Low Presence Low Search, a High Presence High Search and a High Presence Low Search profile. This is partially consistent with our hypothesis and earlier research concerning Meaning in Life-profiles (Dezutter et al., 2013; Dezutter et al.,2014). However, a fourth Low Presence High Search cluster did not appear in both samples. There are several possible explanations for this finding. First, it might represent a meaningful difference, indicating that older adults compared to other populations (i.e., emerging adults and chronically ill adults), when they lack a sense of Presence of Meaning, stop searching. This seems a detrimental path, suggesting that these older adults lost courage to achieve Meaning in Life. A second possible explanation is our study design. The samples in the present study are not representative for the general population of older adults and seemed to represent higher functioning subgroups within residential as well as a community-dwelling populations. The latter is indicated by the low occurrence of depressive symptoms in both samples, compared to the generally high prevalence of depressive symptoms in populations of both residential and community-dwelling older adults (Djernes, 2006). In particular within the residential population, the mean level for depressive symptoms was low (Tsai et al., 2005). Probably a selection bias exists in this study, by only interviewing the higher functioning older adults, who in accordance experienced less depressive symptoms. In addition, the high prescription and use of antidepressants within these settings could be an influential factor, suggesting participants already received treatment against their depressive symptoms.

With regard to our second objective, we analyzed the relationship between the distinct profiles and depressive symptoms. Based on the results, older adults who experienced meaning in their life and did not search, reflected lower levels of depressive symptoms than their counterparts who did not experience meaning and did not search for it. Participants within this cluster of High Presence and Low Search seemed to hold the most beneficial position regarding depressive symptoms. Moreover, this was noticeable in both populations. Furthermore, individuals with a Low Presence Low Search profile, reported higher levels of depressive symptoms regardless of living at home or in residential care. Thus, consistent with Frankl (1968), when older adults do not experience meaning, psychological distress seems to increase, as reflected in higher levels of depressive symptoms.

However, for the third cluster High Presence High Search, this conclusion is less straightforward. Within the residential sample, this cluster non-significantly scored in-between the two other clusters on the level of depressive symptoms. Therefore, this cluster, as the only cluster with a High Search component, did not show a unique relation with depressive symptoms. This is not in line with earlier research in emerging adults and chronically ill adults, that pointed towards a High Search component as a vulnerability for the experience of depressive symptoms (Dezutter et al., 2013; Dezutter et al., 2014). Within the community-dwelling population though, participants with a High Presence High Search profile did not scored in-between: Older adults in the High Presence Low Search cluster and the High Presence High Search cluster reported similar depressive symptoms levels. Thus, in this group, both profiles with High Presence, whether or not combined with High or Low Search, reported lower levels of depressive symptoms. In the residential population, the profile with High Presence, High Search is related with both other profiles. However, in both populations, the component Search for Meaning seems secondary to differentiate between levels of depressive symptoms, in contrast to Frankl (1968) who considered search as the main point of action.

Limitations and Future Directions

Although this study offered a first insight in the complex dynamics between Presence of Meaning and Search for Meaning and their association with late life depressive symptoms, several limitations have to be taken into account. Firstly, this study is done in two homogeneous samples, consisting of mostly Catholic, Flemish and female middle class older adults, therefore it was impossible to generalize findings. In addition, cluster analysis is a data-driven procedure. Future research should replicate these findings, moreover in diverse samples (e.g., older adults with a different religious background). We addressed part of this limitation by comparing our findings for a residential population with a second sample of community-dwelling older adults.

Secondly, this study used self-reported data, thus based on subjective assessments. Additional methods, for example multiple informants methods, could offer more objective information. Above this, different people administered the questionnaires, which may create differences between the scorings, due to interpretation of the scale or the interpretation of the responses of the older adults. A clear guideline for interpretation could offer a partial solution.

Thirdly, The CES-D (Radloff, 1977) is a questionnaire used to assess the whole population and not just older adults. In response to this, we optioned for a shorter format which was validated within a population of older adults (Kohout et al., 1993). Nevertheless, some questions remained difficult due to possible wrong understanding within the (physical) context (e.g., 'During the past week, I could not get "going"'). Another possibility was that the symbolic nature of some questions posed difficulties to understand. Future research could use the Geriatric Depression Scale (Jongenelis et al., 2007), a questionnaire which is especially developed for the assessment of depressive symptoms within a population of older adults. A final limitation was the low mean score on depressive symptoms, as stated above.

Despite these limitations, this study offers a person-oriented view on how Meaning in Life functions within residential and community-dwelling older adults and provides some additional insight into the complex interplay between Presence of Meaning and Search for Meaning, as well as how these constructs play a role in older adults psychological functioning. The present findings affirm the important role of Presence of Meaning for the experience of depressive symptoms and expanded this to the final stage of life. However, further research in this field is necessary in order to replicate the cluster solutions in distinct samples of older adults and to clarify further the role of meaning in the context of late life stressors and processes.

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Tables & Figures (based on 6th APA edition)

Table 1

Theoretical Overview of the Formulated Hypotheses

Expected Cluster	Expected Levels of Depressive Symptoms
High Presence High Search	Low to Moderate
High Presence Low Search	Low
Low Presence High Search	High
Low Presence Low Search	High to Moderate

Table 2

Correlations Among the Study Variables within a population of Residential Older Adults (above the diagonal) and Community-Dwelling Older Adults (below the diagonal)

	Residential Older Adults		
Variable	Depressive symptoms [1.67/.56]	Presence of Meaning [2.92/.77]	Search for Meaning [1.67/.70]
Community-Dwelling Older Adults			
Depressive symptoms [1.67/.48]	-	-.40*	.07
Presence of Meaning [3.20/.58]	-.28*	-	.10
Search for Meaning [2.73/.75]	.03	.08	-

Note. * *p* < .01. Means and standard deviations are between brackets.

Table 3

The Calinski-Harabasz Index for Two to Six Cluster Solutions for the Residential Older Adults and the Community-Dwelling Older Adults

Cluster	Calinski-Harabasz Index for residential older adults	Calinski-Harabasz Index for community-dwelling older adults
Cluster 2	115.550	120.544
Cluster 3	**189.540**	**188.791**
Cluster 4	166.225	164.870
Cluster 5	180.400	177.092
Cluster 6	180.887	189.012

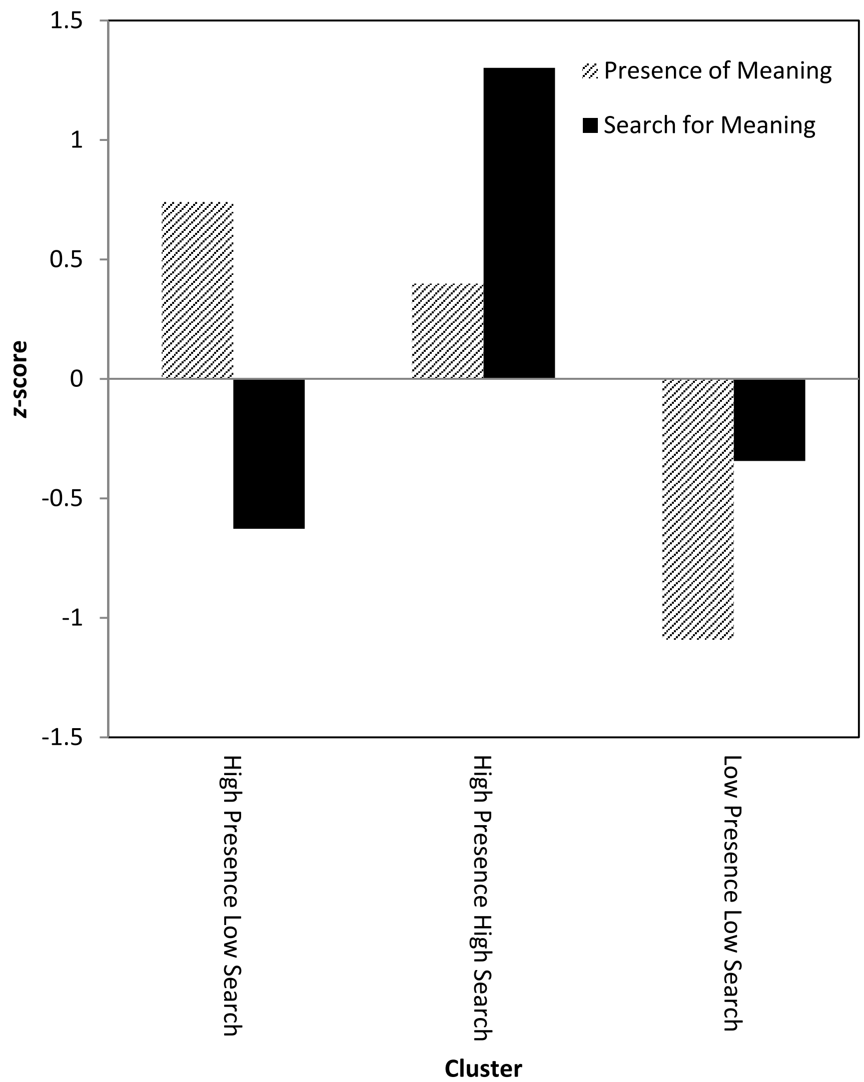

Figure 1. z-scores of Presence of Meaning and Search for Meaning for the Three Clusters within Residential Elderly.

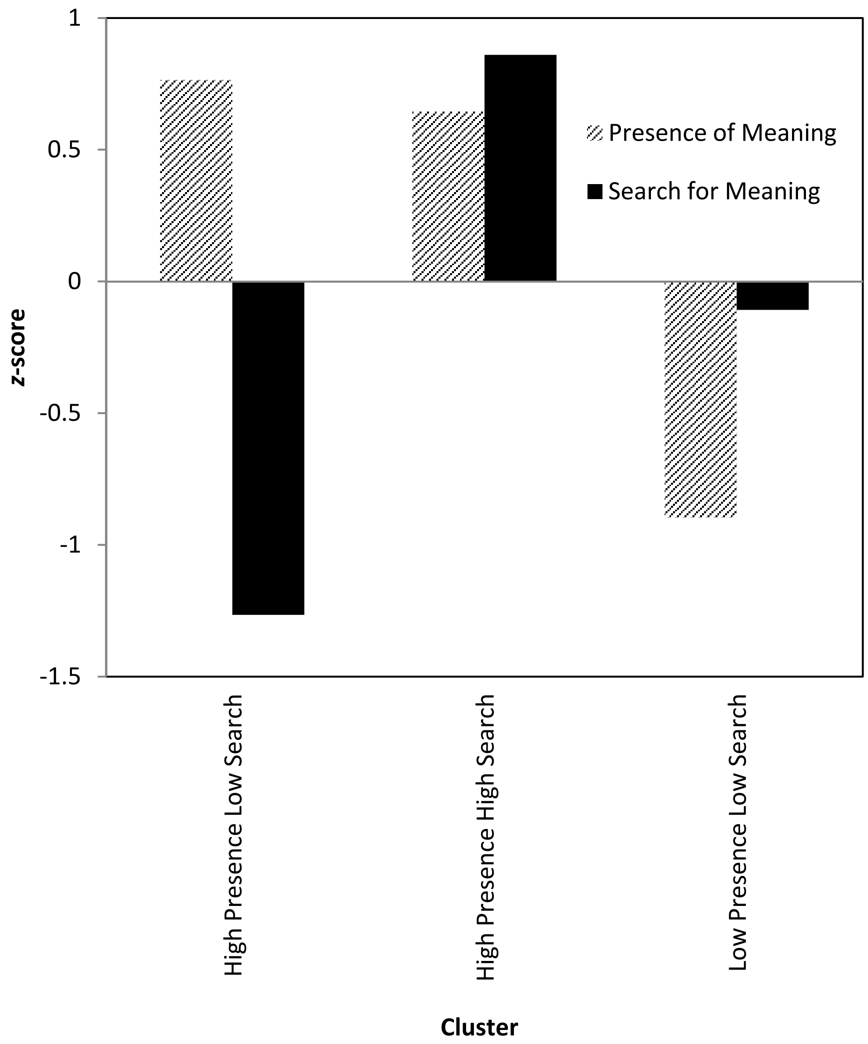

Figure 2. *z*-scores of Presence of Meaning and Search for Meaning for the Three Clusters within Community-dwelling Elderly.